

Report R13 — nested fragmentation: can a rule fragment twice?

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

2026-08-04

Abstract

A rule fragments once when its number of sectors grows exponentially in N . It fragments a *second* time when a single sector contains exponentially many attractors. Only thirteen rules can do the second thing at all — every other rule of the 256 has exactly one attractor per sector (R9 §7.3) — so the question is decidable by inspecting thirteen sequences. The answer is **partly**. Eight rules fragment exponentially *inside* a sector while their sector count stays constant or linear, and that half is settled by exact integer recurrences rather than fits. Three constants appear, all “Narayana-like”: the supergolden $\psi = 1.465571$, the plastic number $\rho = 1.324718$, and the root 1.380278 of $x^4 = x^3 + 1$. The distribution of attractors across sectors for rules 44 and 100 is, exactly, a run of *consecutive Narayana numbers*. The fully nested case — exponential both ways — has two candidates, rules 29 and 71, and **both are left open**. For rule 29, on the even- N branch up to $N = 20$ a quadratic and an exponential of base 1.12 fit its in-sector count to within 6% with essentially equal BIC, and they do not separate by a full attractor until $N \simeq 26$, which is 6.7×10^7 states. Its near-twin rule 71 is exactly linear inside a sector for $N = 6 \dots 18$ and then breaks upward at $N = 19$, so it is open too rather than a clean negative. obc0 only.

1 The two exponents

Every rule has two numbers here:

$n_{\text{wcc}}(N)$ — how many sectors, $m(N) = \max_{\text{sector}} \#\{\text{terminal SCCs in it}\}$ — how many attractors the busiest

$m \equiv 1$ for 243 of the 256 rules, so only the thirteen of R9 §7.3 can contribute. Nested fragmentation means both grow exponentially.

2 Settled: exponential inside a constant or polynomial sector count

rule	tuple	sector count n_{wcc}	attractors in one sector	total base
203, 217	VEII, VIEI	constant, = 1 at every N	exponential ψ	ψ
219	VEEI	constant, = 1	exponential ψ	ψ
36	IDDV	constant, $\rightarrow 10$	exponential ψ	ψ
44, 100	IIDV, IDIV	polynomial (linear)	exponential ψ	ψ
104, 233	DIIV, VIIE	polynomial (linear)	exponential	1.380278
29, 71	EIVD, EVID	exponential	linear then breaks — §3	ρ
235, 249, 37	VEIE, VIEE, EDDV	constant, = 1	constant (≤ 2)	constant

Rules 203 and 217 are the extreme: *one* sector at every N , holding 277 attractors at $N = 16$ and growing like ψ^N . The sector observable returns the constant 1 however large the system

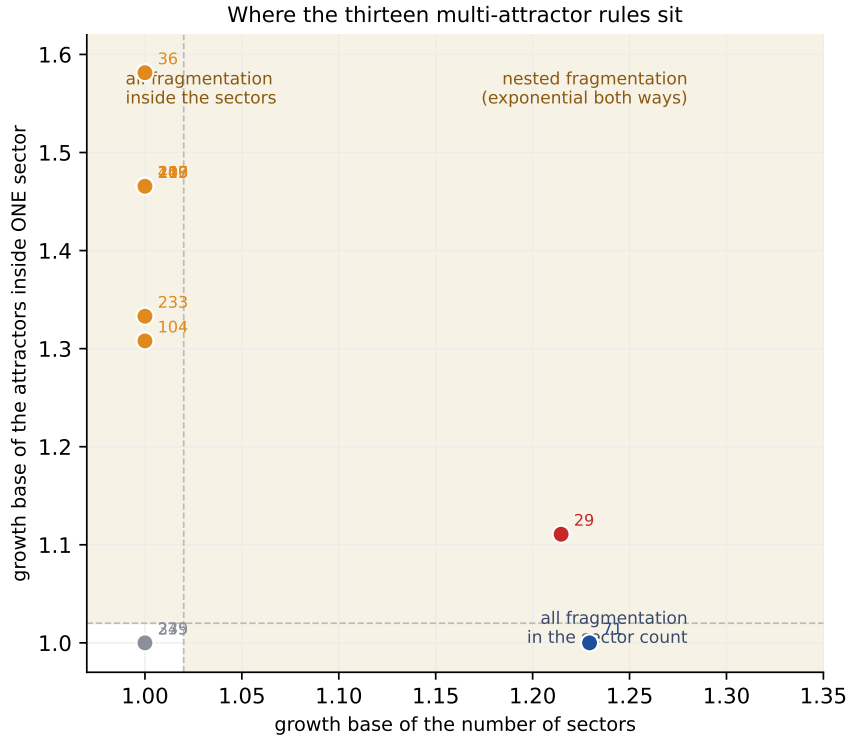


Figure 1: **The two fragmentation exponents.** Horizontal: growth base of the sector count. Vertical: growth base of the attractors inside one sector. The shaded strips are “exponential”. Eight rules sit in the upper-left band — all the fragmentation is inside the sectors — and two (29, 71) in the lower-right — all of it in the sector count. The upper-right corner is nested fragmentation. *Drawn from $N \leq 16$ only:* at that size 29 and 71 look linear inside a sector and sit on the lower-right axis, but §3 shows both break upward by $N = 19$ –20, so their vertical positions here are lower bounds and the corner is not reliably empty.

gets, while the attractor count runs away exponentially — all of the structure is invisible to it. Rules 44 and 100 are the polynomial version: $n_{\text{wcc}} = N + 3$ exactly for 100, with ψ^N attractors inside the largest of those sectors.

2.1 Three exact constants, all Narayana-like

The *total* attractor counts obey exact integer recurrences — found by recurrence search, not fitted:

rules	recurrence	characteristic root	name
203, 217, 219, 36, 44, 100	$a_N = a_{N-1} + a_{N-3}$	$x^3 = x^2 + 1$	supergolden $\psi = 1.465571$
104, 233	$a_N = a_{N-1} + a_{N-4}$	$x^4 = x^3 + 1$	1.380278
29, 71	$a_N = a_{N-2} + a_{N-3}$	$x^3 = x + 1$	plastic $\rho = 1.324718$

The first is Narayana’s cows sequence 1, 1, 1, 2, 3, 4, 6, 9, 13, 19, 28, 41, 60, 88, 129, 189, 277, 406, 595; rules 203/217 enter it at 6, rule 219 at 3, and 36/44/100 at 13. So six rules share one sequence at four different offsets.

2.2 The prettiest fact

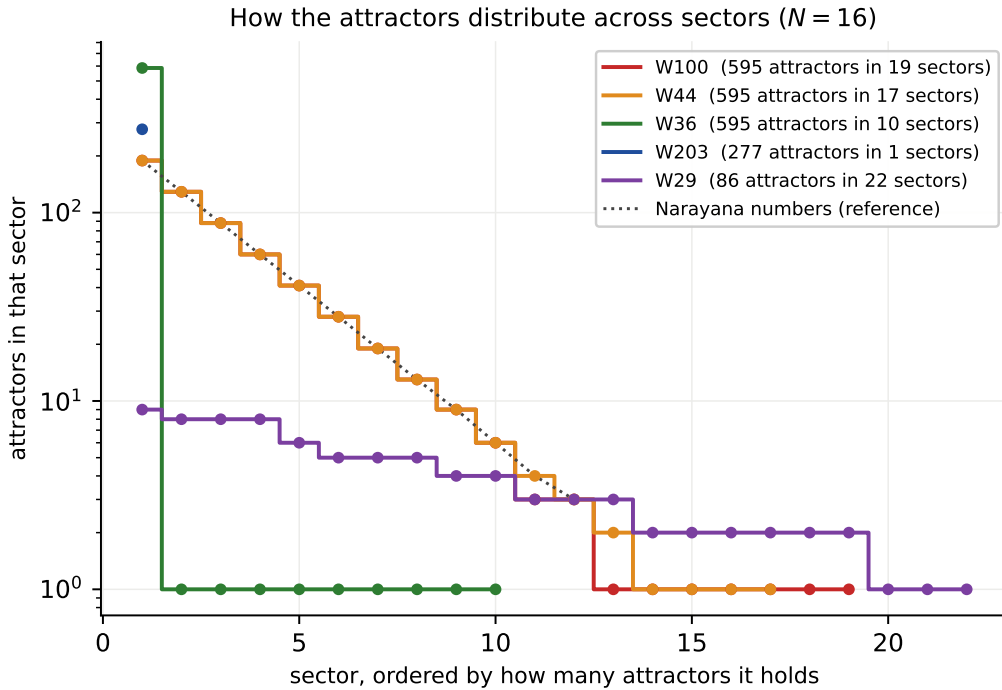


Figure 2: **How the attractors distribute across sectors** ($N = 16$). For rules 44 and 100 the sorted per-sector attractor counts are, exactly, a run of *consecutive Narayana numbers* — the same sequence that gives the total. Rule 36 has the same total, 595, but piles 586 of them into one sector and leaves nine singletons. Rule 203 is a single point: one sector, 277 attractors. Rule 29 spreads 86 attractors thinly over 22 sectors, none holding more than 9.

At $N = 16$ rules 44 and 100 give

$$189, 129, 88, 60, 41, 28, 19, 13, 9, 6, \dots$$

which is the Narayana sequence read backwards from its largest term below the total. Three rules — 36, 44, 100 — have the *identical* total of 595 attractors at $N = 16$ and distribute them completely differently: one giant sector, or a Narayana staircase.

3 Open: the fully nested case

Only 29 and 71 have an exponential sector count, so only they can be nested.

Both are undecided, and in the same way. Each looks exactly linear inside a sector for as far as one first computes, and each then breaks upward at the largest N reached. Rule 71’s in-sector count is

$$1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 6, 7, 7, \mathbf{9} \quad (N = 6 \dots 19),$$

which is $\lceil N/2 \rceil - 2$ *exactly* for $N = 6 \dots 18$ and then returns 9 where the formula says 8. Its even branch 1, 2, 3, 4, 5, 6, 7 is still a perfect line; the break is on the odd branch, at the last point available. Rule 29 breaks earlier and further. So the tidy statement “71 is linear, 29 is not” — which an earlier draft of this report made after stopping at $N = 16$ — is an artefact of stopping too soon.

Rule 29, quantified. Its in-sector count oscillates with parity; the even branch is

$$3, 4, 5, 6, 7, 9, 12, 16 \quad (N = 6, 8, \dots, 20),$$

which is clearly superlinear — a line would give 8 at $N = 18$ and 9 at $N = 20$. But superlinear is not exponential, and the second differences are 0, 0, 0, 1, 1, 1, which is what a *quadratic* looks like. Fitting both:

model	rss	BIC	predicts $N = 22$	predicts $N = 26$
quadratic in N	1.571	−6.78	19.2	28.1
exponential, base 1.1206	1.672	−8.37	18.8	29.7

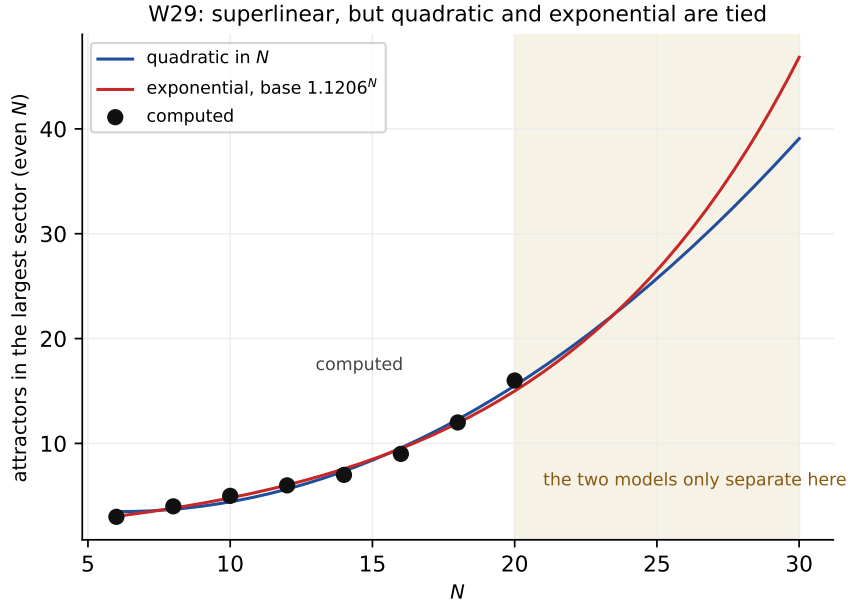


Figure 3: **The undecided case.** Rule 29’s attractors in the largest sector, even N , with both models. They agree to within 6% everywhere data exists and separate by less than one attractor at $N = 22$.

A BIC gap of 1.6 is nothing, and the two models differ by 0.4 attractors at $N = 22$ and 1.6 at $N = 26$. Settling it needs $N \gtrsim 26$, i.e. 6.7×10^7 states — feasible only with a bitwise successor and a compact union-find, not the present path. **Verdict: open.** The automated classifier calls it exponential (base 1.111) and that verdict is recorded, but it rests on a short, parity-oscillating series and should not be quoted as a result.

4 What this says

The eight settled rules answer the second half of the question affirmatively and sharply: *a constant or polynomial sector count is fully compatible with exponentially many attractors inside a single sector*. Rules 203 and 217 are the cleanest statement of it — one sector, forever, containing ψ^N attractors — and they are the reason R9 §7.3 insists that counting sectors is not counting attractors.

Whether a rule can do *both* at once remains open on *two* rules, 29 and 71, by a margin of about one attractor at the largest size computed here. Both are linear inside a sector for a long stretch and both break upward at the end of it, which is the least conclusive shape a finite series can have: it is equally the onset of a slow exponential and a finite-size wobble on a line.

Reproduce

```
python -m qca_fragmentation.viz.nested --bc obc0 --rebuild
```

writes `analytics/nested_fragmentation_obc0.json` and the three figures. The module is `code/qca_fragmentation/viz/nested.py`; `rule29_model_comparison()` reproduces the table of §3.